

CLASS 12 · NCERT ALIGNED · NEET + IIT JEE MAIN

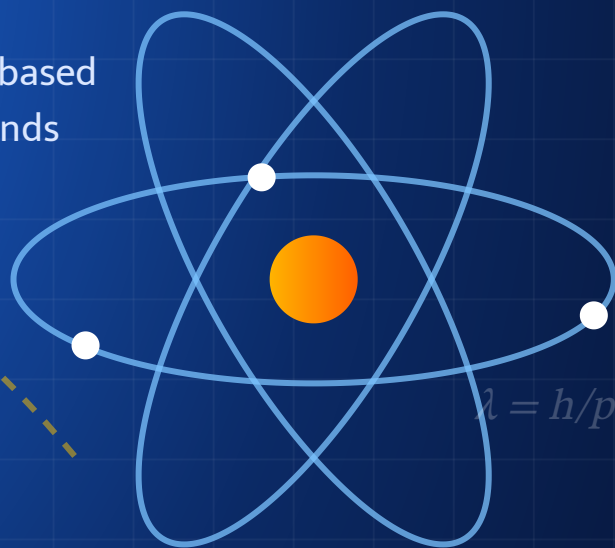
$$1/f = 1/v - 1/u$$

NEET + JEE PHYSICS

30-DAY RAPID REVISION

Concepts · Formulas · Diagrams · Traps · Pattern-based practice from NEET & JEE Main previous-year trends

CLASS 12



$$E = h\nu$$

14 NCERT Chapters

300+ MCQs with Solutions

Full Mock Test

Board + NEET + JEE Main

100+ Diagrams & Formula Boxes

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VS SIR

Vijay Kumar Sharma
20+ years · NEET/JEE Physics · Kota

NEET & JEE Main Physics: 30-Day Rapid Revision (Class 12)

Class 12 Physics · NCERT-aligned · For NEET & IIT JEE Main

by VS Sir (Vijay Kumar Sharma)

AI-Enabled Educator · 20+ years teaching NEET/JEE Physics, Kota

First edition: 2026 · Language: English

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About the questions: All practice questions in this book are original, pattern-based questions written by the author and modelled on the trends of NEET and JEE Main previous-year papers. They are not reproductions of official papers. Syllabus follows the rationalised NCERT Class 12 Physics textbook (Parts I & II), the NMC NEET-UG syllabus and the NTA JEE Main syllabus.

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ABOUT THE AUTHOR

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- 20+ years of teaching Physics to NEET and JEE aspirants in Kota
- AI-Enabled Educator – trains teachers to use AI tools in the classroom
- Author of "NEET Physics: 30-Day Rapid Revision (Class 11)" in English and Hindi, "Electrostatics" and "Practical Guide to AI for Educators"
- Known for turning long chapters into one-page concept maps and formula boxes

A message from VS Sir

Dear student,

In the last month before NEET or JEE Main, you do not need another 800-page book. You need a **clear, short, reliable revision** of everything NCERT says, the formulas that actually get used, and the traps where toppers lose marks.

This book is built exactly for that. One chapter per day or two, a formula box you can revise in five minutes, diagrams that make the idea obvious, and practice questions in the pattern NEET and JEE Main have followed year after year – each with a quick solution.

Follow the 30-day plan honestly, tick each box, and keep an error log. Consistency beats intensity. You can do this!

— VS Sir

How to use this book

Every chapter follows the same rhythm so your brain knows exactly where to look.

1. **Read** the chapter-at-a-glance map (2 min).
2. **Revise** concepts, diagrams and formula boxes (40–60 min).
3. **Solve** the 20 practice questions with a timer – 1 min each.
4. **Check** with the answer key; read the solution of every wrong/guessed question.
5. **Log** each mistake in your error log (page at the end). Re-solve on Day 29.

Colour code of the boxes

 Key Concept Core idea explained simply	 Formula Box Formulas to revise in 5 minutes
 Common Trap Where students lose marks	 VS Sir's Exam Tip Exam strategy & shortcuts
 NCERT Line to Remember Exact NCERT facts asked in NEET	 Memory Trick Tricks to remember
 JEE Main Extension Extra depth for JEE Main	

Question tags

NCERT based direct from textbook lines **NEET pattern** typical NEET level **JEE Main pattern** one step deeper
Assertion–Reason

 Honest note on PYQs The questions are original and written in the style of NEET and JEE Main previous-year questions (PYQ trends): same concepts, same difficulty, same traps. Solve your official PYQ papers too – this book prepares you to crack them.

Syllabus & references

This book covers the complete Class 12 Physics syllabus common to NEET-UG and JEE Main. Topics asked only in JEE Main are marked with a JEE box. Standard books were used only to cross-check results – all explanations, diagrams and questions in this book are original.

NCERT Physics Class XII, Parts I & II (rationalised edition)

Primary reference for concepts, definitions and standard values.

NEET-UG Physics syllabus – National Medical Commission (NMC); exam conducted by NTA · neet.nta.nic.in

Topic coverage and depth for NEET.

JEE Main Physics syllabus – National Testing Agency (NTA) · jeemain.nta.nic.in

Topic coverage and depth for JEE Main; JEE-only extras are marked.

H. C. Verma – Concepts of Physics, Vol. 2

Cross-check of derivations, sign conventions and standard results.

D. C. Pandey – Understanding Physics (Electricity & Magnetism; Optics & Modern Physics)

Cross-check of problem types asked in NEET and JEE Main.

Always check the latest official syllabus

NTA and NMC publish the syllabus and information bulletin for every exam year. Before your exam, confirm the current syllabus on the official websites.

Your 30-Day Revision Plan

Daily target: 2–3 hours for Physics. Tick the box when a day is complete. Weekly tests are built into Days 8 and 17.

Day 1 Ch 1 Charges, Coulomb, electric field	Day 2 Ch 1 Dipole, flux, Gauss's law	Day 3 Ch 2 Potential, equipotentials, energy	Day 4 Ch 2 Capacitors, dielectrics	Day 5 Ch 3 Drift, Ohm's law, resistivity
Day 6 Ch 3 Cells, Kirchhoff, Wheatstone	Day 7 Ch 4 Biot–Savart, Ampere's law	Day 8 Ch 4 Forces, galvanometer + Test (Ch 1–4)	Day 9 Ch 5 Magnetism and matter	Day 10 Ch 6 Electromagnetic induction
Day 11 Ch 7 AC: phasors, R, L, C circuits	Day 12 Ch 7 LCR, resonance, power, transformer	Day 13 Ch 8 Electromagnetic waves	Day 14 Ch 9 Reflection, refraction, TIR	Day 15 Ch 9 Lenses, prism, instruments
Day 16 Ch 10 Huygens, interference, YDSE	Day 17 Ch 10 Diffraction, polarisation + Test (Ch 5–10)	Day 18 Ch 11 Photoelectric effect, de Broglie	Day 19 Ch 12 Atoms: Bohr model, spectra	Day 20 Ch 13 Nuclei: binding energy, decay
Day 21 Ch 14 Semiconductors, p–n junction	Day 22 Ch 14 Diode circuits, rectifiers, logic gates	Day 23 Final Formula sheet: Electricity & Magnetism	Day 24 Final Formula sheet: Optics & Modern Physics	Day 25 Final Experimental skills (NEET/JEE)
Day 26 Final Traps list + important graphs	Day 27 Final FULL MOCK TEST (45 Q, 60 min)	Day 28 Final Mock analysis + error log	Day 29 Final Re-attempt all wrong questions	Day 30 Final Exam-day plan + final revision

Golden rules

- Never skip the practice set – revision without questions is only reading.
- Sleep 7 hours. Memory is stored while you sleep.

If you fall behind

Do the formula box + 10 questions of the missed day, then continue with today. Use Days 28–29 as buffer.

Contents

1	Electric Charges and Fields Charge & quantisation · Coulomb's law · Electric field · Electric dipole	Day 1–2	8
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★	Last-Week Toolkit: Formulas, Experiments, Traps & Strategy 14 formula sheets · 9 experiments · 50 traps · 14 key graphs · exam-day plan · error log	Day 23–26	144
✍	Full Mock Test 45 questions · 60 minutes · NEET marking · Class 12 full syllabus	Day 27–29	162

CHAPTER

1

Electric Charges and Fields

17

Day 1-2

Weightage ★★★

NCERT Class 12

Charge & quantisation

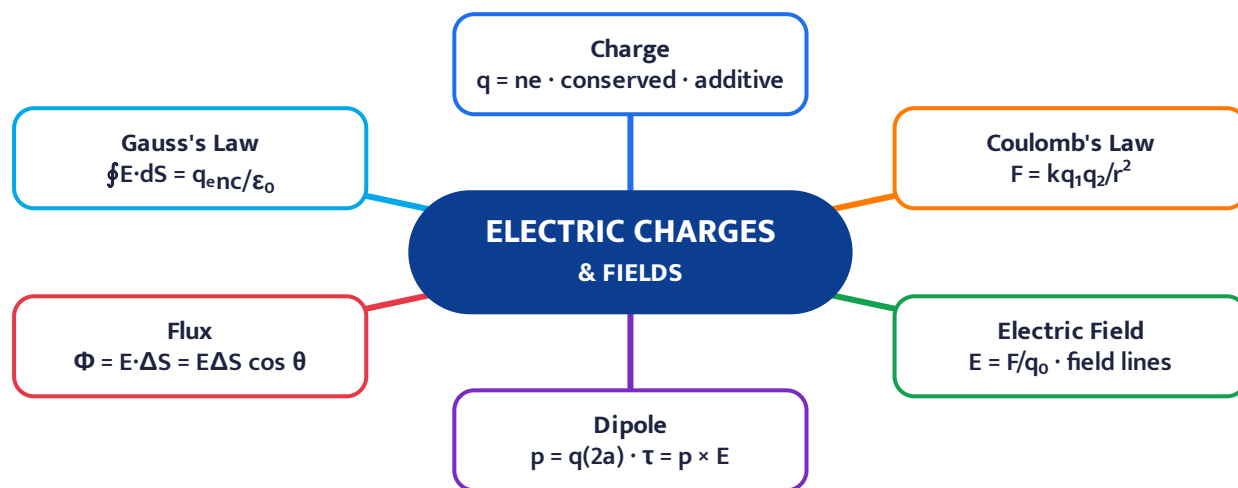
Coulomb's law

Electric field

Electric dipole

Electric flux

Gauss's law



Chapter at a glance

1.1 Electric Charge and its Properties

Electric charge is an intrinsic property of matter due to which it experiences electric and magnetic forces. There are two kinds – **positive** and **negative**; like charges repel, unlike charges attract. Charging by rubbing happens by **transfer of electrons**: the body that loses electrons becomes positive. SI unit: **coulomb (C)**.

3 basic properties

- **Additivity** – total charge = algebraic sum:
 $q = q_1 + q_2 + \dots$ (charge is a scalar).
- **Conservation** – total charge of an isolated system never changes; charge is neither created nor destroyed.
- **Quantisation** – $q = \pm ne, n = 0, 1, 2, \dots$;
 $e = 1.6 \times 10^{-19} \text{ C}$.

NCERT Line to Remember

- At the macroscopic level (charges of μC) quantisation can be ignored and charge treated as **continuous**.
- $1 \text{ C} \approx 6.25 \times 10^{18}$ electrons. Mass is transferred too, but negligibly.
- A charge is always attached to mass; charge is invariant with speed.

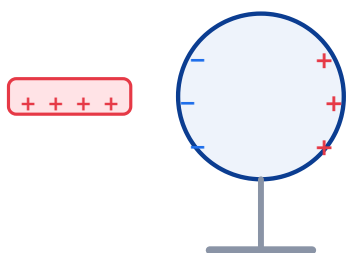
Charging by rubbing (NCERT pairs)

Glass rod rubbed with silk → glass +, silk –. Plastic/ebonite rod rubbed with wool or fur → rod –, wool +. Both bodies always get **equal and opposite** charges.

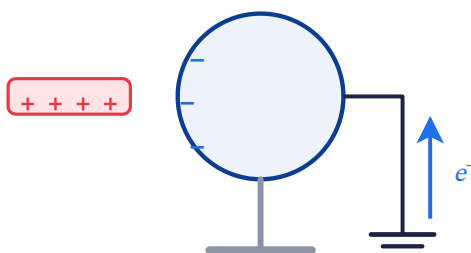
1.2 Conductors, Insulators and Charging by Induction

Conductors (metals, human body, earth) have free electrons and let charge move; charge given to a conductor spreads over its **outer surface**. **Insulators** (glass, plastic, nylon) hold charge where it is placed. **Earthing** (grounding) shares charge with the huge earth, so the body is neutralised. In **induction**, a charged body rearranges charges in a nearby conductor **without touching** it – the conductor finally gets a charge **opposite** to the inducing charge.

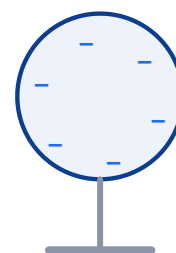
1. Bring a +ve rod near



2. Earth the far side



3. Remove earth, then rod



sphere gets opposite (-) charge

Charging a metal sphere by induction – the rod never touches the sphere

! Common Trap

- A charged body attracts a **neutral** body too (induced opposite charge sits nearer). So **attraction does not prove** the other body is charged – **repulsion is the sure test**.
- Remove the earth connection **before** removing the rod; otherwise the induced charge flows back and the sphere stays neutral.

1.3 Coulomb's Law and Superposition

The force between two stationary point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, acting **along the line joining them**.

! Coulomb's law

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

$$k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}, \epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\vec{F}_{21} = k \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}, \quad \vec{F}_{12} = -\vec{F}_{21}$$

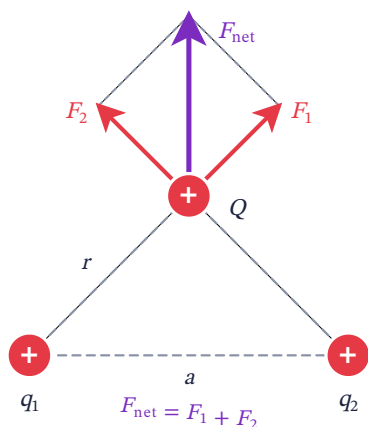
Vector form: force on 2 due to 1; $\hat{r}_{21}$ points from 1 to 2 (obeys Newton's third law)

$$F_{\text{medium}} = \frac{F_{\text{vacuum}}}{K}, \quad K = \frac{\epsilon}{\epsilon_0} = \epsilon_r$$

K = dielectric constant (relative permittivity) of the medium

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1n}$$

Superposition: each pair acts independently; forces add as vectors



Superposition of Coulomb forces

Solved Example

Charges $q_1 = q_2 = +1 \mu\text{C}$ sit at A and B, 0.2 m apart. A charge $Q = +1 \mu\text{C}$ is at a point 0.1 m directly above the midpoint of AB. Find the net force on Q.

Solution: $r = \sqrt{0.1^2 + 0.1^2} = 0.1\sqrt{2} \text{ m}$,
 $F_1 = F_2 = \frac{9 \times 10^9 \times 10^{-12}}{0.02} = 0.45 \text{ N}$ at 45° to the vertical.
 Horizontal parts cancel: $F_{\text{net}} = 2(0.45)\cos 45^\circ \approx \mathbf{0.64 \text{ N}}$,
vertically up.

Common Trap

- Coulomb's law is for **point** charges at rest. Force between two charges is **unchanged** by a third charge nearby (superposition) – only the net force changes.
- Touching two identical conducting spheres makes each charge $\frac{q_1 + q_2}{2}$ (with signs!).

Memory Trick

Electric vs gravitational force between two protons: $F_e / F_g \approx 10^{36}$. Electric force can attract **or** repel; gravity only attracts. Both obey the inverse-square law and are central forces.

1.4 Electric Field and Field Lines

The **electric field** at a point is the force per unit positive test charge placed there: $\vec{E} = \lim_{q_0 \rightarrow 0} \vec{F} / q_0$ (the test charge must be tiny so it does not disturb the source). Unit: $\text{N C}^{-1} = \text{V m}^{-1}$. A charge q in a field feels $\vec{F} = q\vec{E}$ – a positive charge along $\vec{E}$, a negative charge (electron) **opposite** to $\vec{E}$.

Electric field

$$\vec{E} = k \frac{q}{r^2} \hat{r}$$

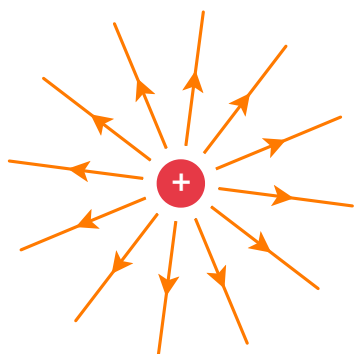
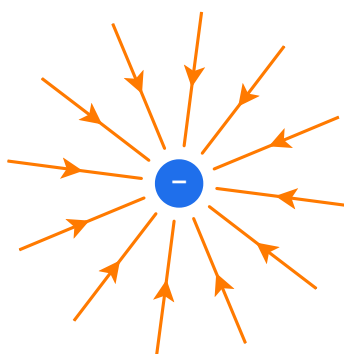
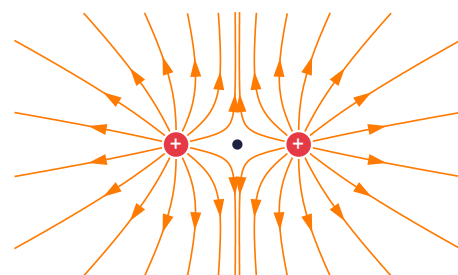
Point charge: radially outward for $+q$, inward for $-q$; $E \propto 1/r^2$

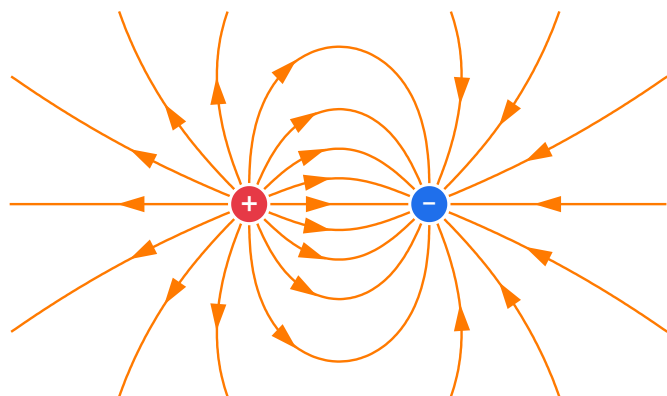
$$\vec{E} = \sum_i k \frac{q_i}{r_i^2} \hat{r}_i$$

System of charges – superposition

$$a = \frac{qE}{m}$$

Acceleration of a charged particle in a uniform field (path parabolic if it enters $\perp$ to E)

Isolated $+q$: lines go outIsolated $-q$: lines come inTwo equal $+q$: neutral point (dot) midway



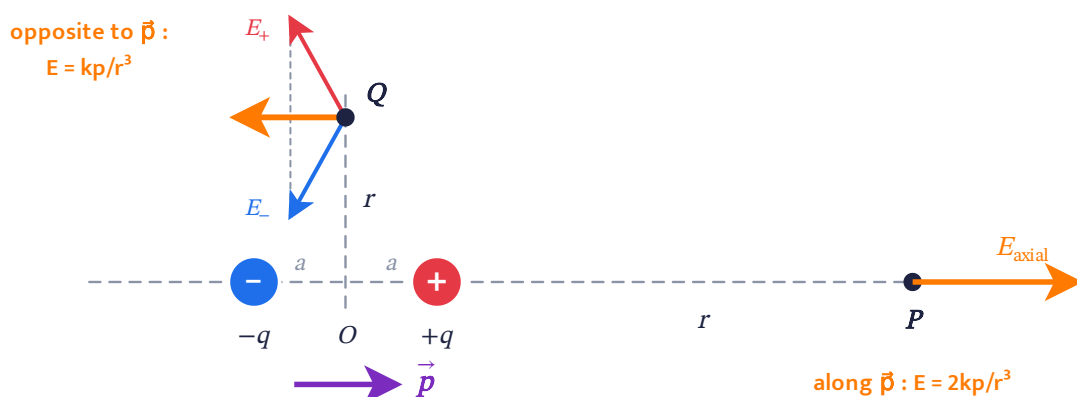
Electric dipole (+q, -q): lines start on +q and end on -q

Properties of field lines

- Start on +ve charge, end on -ve charge (or at infinity).
- Continuous curves in a charge-free region; **never form closed loops** (field is conservative).
- **Two lines never cross** – E has one direction at a point.
- Tangent gives direction of E; **crowding** shows strength.

1.5 Electric Dipole

Two equal and opposite charges $\pm q$ separated by a small distance $2a$ form an **electric dipole**. Its **dipole moment** $\vec{p} = q(2\vec{a})$ points **from -q to +q** (unit C m). Total charge is zero, but the field is not – it falls as $1/r^3$, faster than a point charge. Polar molecules such as H_2O have a permanent dipole moment.



Field of a dipole at an axial point P and an equatorial point Q

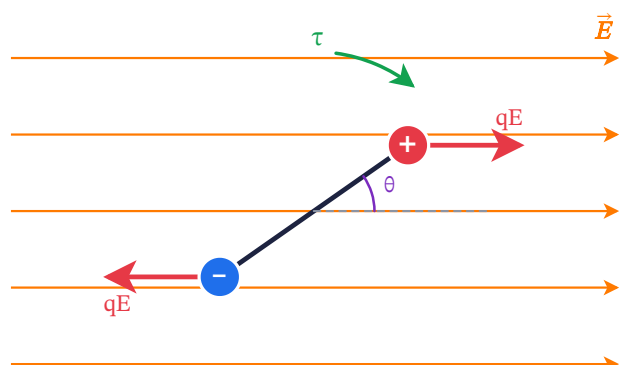
Dipole Formulas

$$E_{\text{axial}} = \frac{2kpr}{(r^2 - a^2)^2} \approx \frac{2kp}{r^3} \quad \text{On the axis, along } \vec{p} \text{ (for } r \gg a)$$

$$E_{\text{eq}} = \frac{kp}{(r^2 + a^2)^{3/2}} \approx \frac{kp}{r^3} \quad \text{On the equatorial plane, opposite to } \vec{p}$$

$$E_{\text{axial}} : E_{\text{eq}} = 2:1 \quad \text{At the same large distance}$$

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta \quad \text{Torque in a uniform field; net force = 0}$$



Net force = 0; $\tau = pE \sin \theta$ turns $\vec{p}$ towards $\vec{E}$

Dipole in a uniform field – a couple acts

Solved Example

A dipole of charges $\pm 10 \mu\text{C}$ separated by 5 mm. Find E at 15 cm from its centre on (i) the axis, (ii) the equatorial line.

Solution: $p = 10^{-5} \times 5 \times 10^{-3} = 5 \times 10^{-8} \text{ C m}$. (i)

$$E = \frac{2kp}{r^3} = \frac{2(9 \times 10^9)(5 \times 10^{-8})}{(0.15)^3} \approx 2.67 \times 10^5 \text{ N/C along } \vec{p}.$$

Half: $1.33 \times 10^5 \text{ N/C}$, opposite to $\vec{p}$.

VS Sir's Exam Tip

- Uniform field $\Rightarrow \mathbf{F} = \mathbf{0}$, $\tau \neq \mathbf{0}$ (only rotation). Non-uniform field $\Rightarrow$ a net force also acts; when $\vec{p} \parallel \vec{E}$ the dipole moves towards the **stronger** field.
- τ is maximum ($= pE$) at $\theta = 90^\circ$; zero at $\theta = 0^\circ$ (**stable**) and 180° (**unstable** equilibrium).

1.6 Continuous Charge Distributions

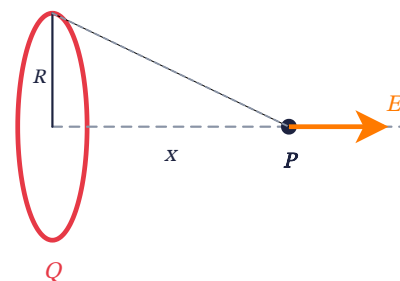
Distribution	Density	Unit	Element
Line (wire, ring)	$\lambda = \frac{dq}{dl}$	C m^{-1}	$dq = \lambda dl$
Surface (sheet, shell)	$\sigma = \frac{dq}{dS}$	C m^{-2}	$dq = \sigma dS$
Volume (solid sphere)	$\rho = \frac{dq}{dV}$	C m^{-3}	$dq = \rho dV$

Field of a continuous body: split it into elements dq , treat each as a point charge and add (integrate) vectorially:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$

JEE Main Extension

- Ring (charge Q , radius R) on its axis: $E = \frac{kQx}{(R^2 + x^2)^{3/2}}$; $E = 0$ at centre, **maximum** at $x = R / \sqrt{2}$; for $x \gg R$ it acts like a point charge.

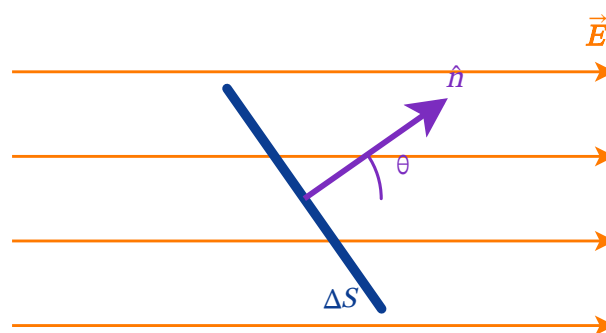


Uniformly charged ring – field on the axis

1.7 Electric Flux

Electric flux through a surface measures the number of field lines crossing it. For a small area ΔS with outward normal $\hat{n}$: $\Delta\Phi = \vec{E} \cdot \vec{\Delta S} = E \Delta S \cos\theta$. It is a **scalar**; unit $\text{N m}^2 \text{C}^{-1}$ ($= \text{V m}$).

- $\theta = 0^\circ$: $\Phi = E\Delta S$ (maximum); $\theta = 90^\circ$: $\Phi = 0$ (field grazes the surface).
- Closed surface: outward flux +ve, inward flux -ve.
- Uniform $\vec{E}$ through any closed surface: net flux = 0.



$$\Phi = E \Delta S \cos \theta \quad (\theta: \text{between } \vec{E} \text{ and normal})$$

Flux through a tilted area

1.8 Gauss's Law and its Applications

The total flux through **any closed surface** equals $1 / \epsilon_0$ times the net charge enclosed. Choose a **Gaussian surface** that matches the symmetry so that E is constant on it and $\vec{E} \parallel d\vec{S}$ (or $\perp$) – then $E \times (\text{area}) = q_{\text{enc}} / \epsilon_0$.

$$1/f = 1/v - 1/u$$

All the best for NEET & JEE!

$$F = kq_1q_2/r^2$$

Revise daily, test weekly, analyse every mistake. Physics rewards the student who practises smartly.

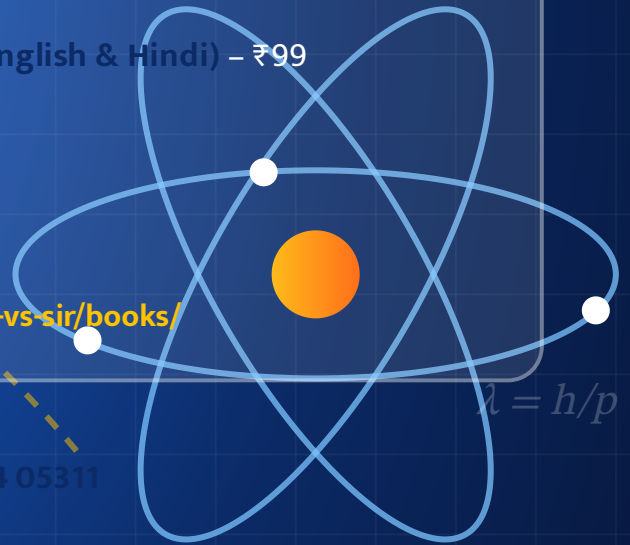
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- Electrostatics (NEET/JEE) – ₹49
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$$E = h\nu$$



$$\lambda = h/p$$

$$V = IR$$

$$\varepsilon = -d\Phi/dt$$

